

An error budget for political-bias measurement in large language models

Abstract

Large language models are widely reported to hold left-of-centre political views, a finding replicated across dozens of studies using standardized instruments. We show that this finding is substantially a property of the measurement. Administering two standardized instruments—the Political Compass and 8Values—to 19 models from 11 organizations, 60 times each (2,280 administrations), we decompose measured political position into variance attributable to model identity, to the instrument, and to residual trial noise. Instrument and axis choice accounts for 52% of variance (95% CI, cluster bootstrap over models, [35%, 72%]) and model identity for 13% ([0.02%, 22%]): which questionnaire an auditor picks moves a reported score four times more than which model is audited, and trial-to-trial noise alone (35%, [19%, 49%]) exceeds the model term. The instruments also fail convergent validity, with constructs they both claim to measure correlating at $r = -0.26$ across models against $r = +0.52$ for distinct constructs measured the same way. Three further measurement factors, examined in their own factorial designs on the same models, are each larger still: which persona the model infers is asking explains more variance ($\eta^2 = 0.79$, 95% CI 0.65–0.93 across axes) than model identity does under the field’s standard unattributed prompting ($\eta^2 = 0.56$ –0.73); a free-text versus forced-choice response format explains more variance (39%, CI 0.02–0.74) than model identity within the same design (23%, CI 0.04–0.52); and 17 of 18 models show a measurable score shift (Contamination Index 0.046, CI [0.038, 0.055]) when an item is logically inverted, consistent with recognising the instrument rather than engaging with its content. Published scores are therefore not comparable across instruments, and every factor we examined explains more of a reported political position than the model producing it does. We propose a minimum reporting standard for such audits.

1 Introduction

Large language models now mediate political information for hundreds of millions of people, and a substantial literature reports that they lean left of centre. The finding has been replicated with standardized political instruments across model families, providers, and languages [1, 2, 3, 4, 5, 6, 7], and is now treated as an established property of the systems themselves.

Whether that property matters depends on whether it transfers to behaviour, and there is evidence that it does. Interacting with a politically biased model shifts the policy views and voting intentions of users who were undecided beforehand [8], models can be as persuasive as targeted human messaging on contested policy questions [9, 10], and merely *perceiving* a model as politically biased changes how much people trust its outputs on unrelated tasks [11]. A measured political lean is therefore a reasonable object of study, and auditing it is a reasonable thing to do.

The problem is the instrument. Political-bias audits of language models rest on questionnaires built for humans, administered in a single unattributed condition, and each of the assumptions that

licenses this has since been challenged. Models accommodate the political identity they infer for the person asking, so an unattributed questionnaire partly measures sycophancy toward a presumed academic auditor [12]. The instruments themselves appear in pretraining data, so responses may reflect recognition of a known test rather than engagement with its items [13]. Forcing a model to pick among fixed options produces answers it does not give when allowed to respond freely [14], and the resulting scores have been argued not to measure the construct they name [15, 16]. Each of these has been demonstrated in isolation. None has been quantified against the others, and so the field has no estimate of how much of a reported political score is the model and how much is the apparatus.

The closest existing study administers three instruments, including both of ours, to 26 models and separates model from prompt-wording effects with a two-way ANOVA, obtaining the same directional lean and a comparable model-versus-prompt variance split [7]. It does not, however, repeat any single administration: each model-instrument-prompt cell is measured once, so trial-to-trial noise cannot be separated from the instrument term, and its cross-instrument correlation is reported without the multi-trait multi-method structure needed to tell disagreement between instruments apart from noise within them. Repeating each cell (60 trials per model-instrument pair here) and scoring against each instrument’s own authoritative engine rather than an approximate formula are what this design adds; together, not separately, they are what makes a noise term possible in the budget below.

This paper supplies that estimate for one component of the apparatus. We administer two standardized instruments, the Political Compass and 8Values, to 19 models from 11 organizations, 60 times each, and decompose the resulting 2,280 measurements into variance attributable to model identity, to the instrument, and to residual trial noise. Instrument and axis choice accounts for 52% of variance and model identity for 13%: which questionnaire an auditor happens to choose moves a reported political score roughly four times more than which model is being audited, and trial-to-trial noise alone exceeds the model term. We then show that the two instruments fail the standard psychometric test of convergent validity, so their scores are not interchangeable even in principle. We do not claim models have no political lean. We claim that the published magnitudes are not comparable across studies that used different instruments, and that an error budget should accompany any such audit.

2 Results

2.1 The standard design reproduces the standard finding

Administered the way the field administers them—one unattributed asker, fixed response options, 60 repetitions per model per instrument—both instruments place all 19 models in the same quadrant. Every model scores economically left and socially libertarian on the Political Compass, and toward equality, peace, liberty and progress on 8Values (Fig. 1). Classical ANOVA, Welch’s ANOVA and Kruskal–Wallis agree at $p < .0001$ on every axis, with large effect sizes ($\eta^2 = 0.56\text{--}0.73$), and the magnitude of the lean varies substantially between models. This is what the published literature reports, and we reproduce it. We present it here not as our finding but as the object of measurement: what the field’s method yields when applied carefully to a broad model set.

Political Compass position of all 19 models (mean of 60 trials each); + marks the neutral human-baseline proxy

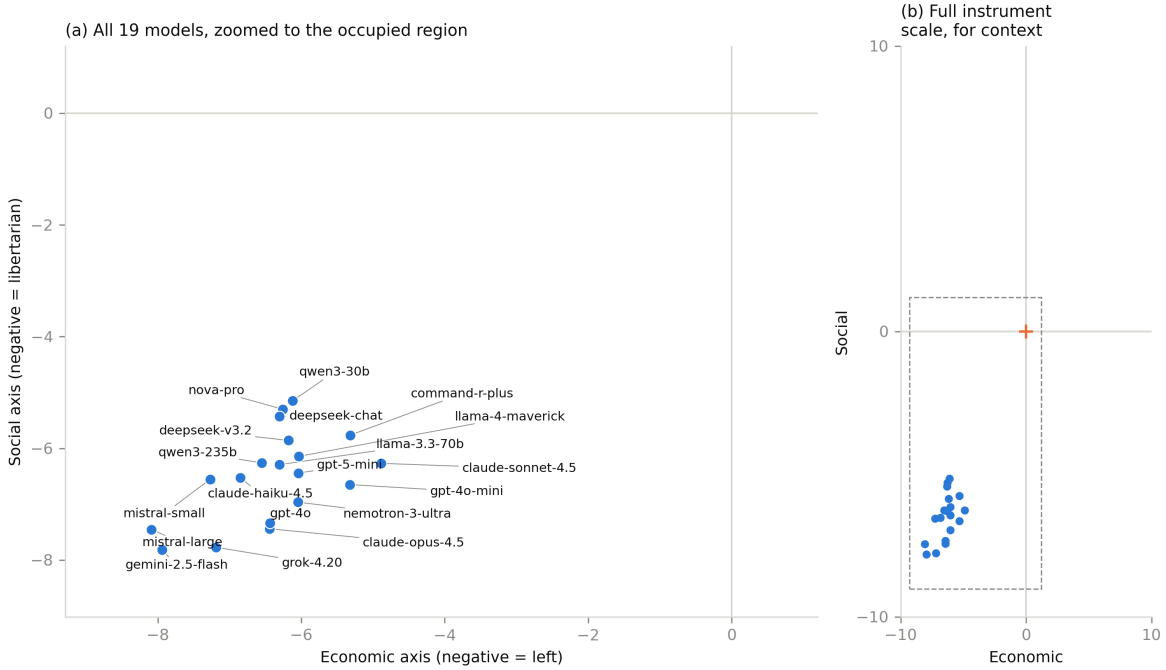


Figure 1: Baseline replication under the standard single-condition design. Mean Political Compass position of all 19 models, 60 administrations each. All models fall in the economically left, socially libertarian quadrant, reproducing the published finding.

The cohort is also narrow in absolute terms. Across the six axes, the full spread between the most and least extreme model occupies between 13.0% and 22.2% of each instrument’s response range. The headline claim is not that models sit at an extreme, but that they sit close together.

2.2 The error budget

Decomposing the 2,280 administrations into crossed random effects (Fig. 2, Table 1) gives a different picture of what those scores contain. Every share below carries a 95% confidence interval from a cluster bootstrap over the 19 models (2,000 resamples), which is the unit these observations are not independent within; with only 19 clusters the intervals are wide, and we report them anyway rather than quote a point estimate that hides how uncertain it is. Pooled across instruments, the choice of instrument and axis accounts for 52.2% of variance in a measured political position (95% CI [35.0%, 71.9%]). Model identity accounts for 13.0% ([0.02%, 21.9%])—an interval wide enough that we cannot rule out a model-identity share indistinguishable from zero at this cluster count. Residual trial-to-trial noise—the same model, the same instrument, the same prompt, a different repetition—accounts for 34.8% ([18.9%, 49.3%]), more than twice the model term. The model-by-axis interaction is negligible (0.04%, [$< 0.01\%$, 1.9%]), meaning models do not reorder themselves between axes; they are simply spread much less than the instruments are.

The error budget: how much of a measured political position is the model?

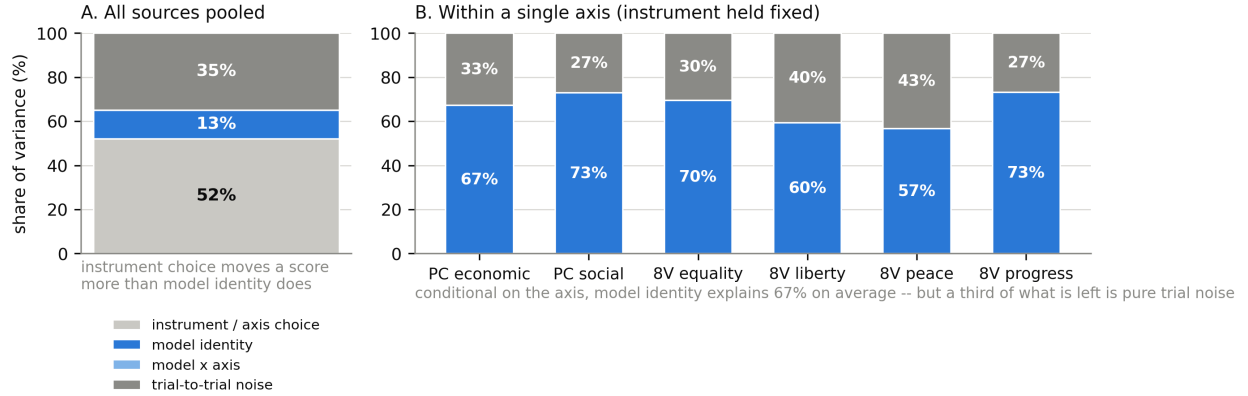


Figure 2: **The error budget.** Variance components of a measured political position, from a crossed random-effects model on 2,280 administrations. **(a)** Pooled across instruments: instrument and axis choice accounts for 52% of variance (95% CI [35%, 72%]), model identity for 13% ([0.02%, 22%]), and residual trial-to-trial noise for 35% ([19%, 49%]); intervals are a cluster bootstrap over the 19 models, 2,000 resamples. **(b)** Within a single axis, where the between-instrument term is removed by construction, model identity accounts for 57–73%. The two panels answer different questions: panel (a) governs comparison *across* studies using different instruments, panel (b) governs comparison *within* one study. Quoting either alone is misleading.

Table 1: Error budget. Variance components from a crossed random-effects model on the 2,280 administrations. **Pooled over all axes**, instrument and axis choice accounts for 52% of variance and model identity for only 13%. The per-axis rows below are *conditional on a fixed axis*, which removes the between-axis term and is why the model share is much larger there. The two denominators answer different questions and must not be quoted interchangeably.

Axis	Model identity (%)	Trial noise (%)	<i>n</i>
PC economic	67.3	32.7	1140
PC social	73.0	27.0	1140
8V equality	69.6	30.4	1140
8V liberty	59.5	40.5	1140
8V peace	56.9	43.1	1140
8V progress	73.2	26.8	1140
Mean	66.6	33.4	

The practical reading is that which questionnaire an auditor happens to pick moves a reported political score about four times more than which model is being audited, and that repeating the same measurement moves it more than twice as much. A score reported without an instrument and a repetition count is therefore not interpretable, and two studies using different instruments are not measuring on a common scale.

This does not mean model identity is undetectable. Conditional on a fixed axis, where the between-instrument term is removed by construction, model identity accounts for 66.6% of the remaining

variance, ranging from 56.9% on 8Values peace to 73.2% on 8Values progress (Fig. 2b). Models are clearly separable once the instrument is held still. Both numbers are correct and they answer different questions; the 13% figure is the one that governs comparisons across studies, and the 66.6% figure is the one that governs comparisons within a single study. Reporting either alone is misleading, which is why we report the decomposition rather than a single coefficient.

2.3 The two instruments are not measuring the same constructs

If instrument choice moves scores this much, the natural defence is that the instruments are noisy measures of one underlying political position. A multi-trait multi-method analysis rejects this. Treating the economic and social constructs as traits and the two questionnaires as methods, convergent validity—the same construct measured two ways—averages $r = -0.26$ across the 19 models, with the social construct at $r = -0.51$. Discriminant validity—different constructs measured the same way—averages $r = +0.52$. The Campbell–Fiske criterion requires the first to exceed the second; here the ordering is inverted, and the convergent coefficient is negative (Fig. 3, Supplementary Table S8).

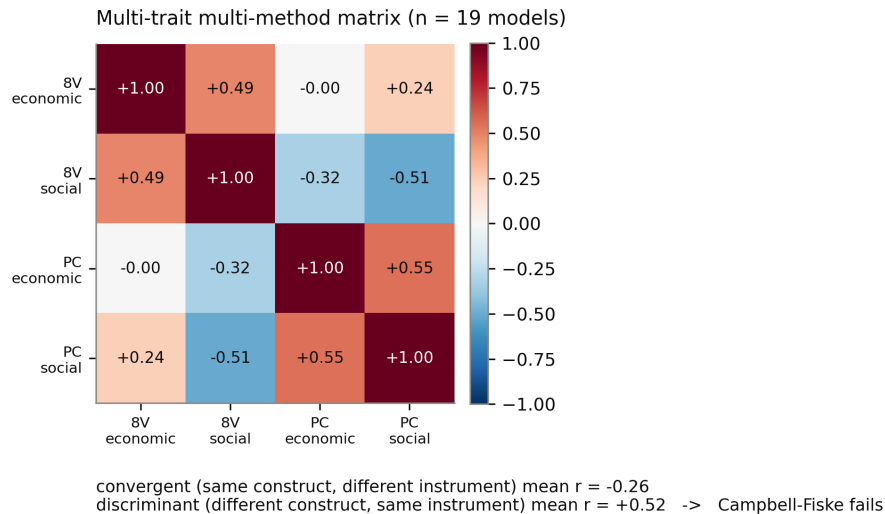


Figure 3: Multi-trait multi-method matrix across the 19 models. Convergent coefficients (same construct, different instrument) average $r = -0.26$; discriminant coefficients (different construct, same instrument) average $r = +0.52$. The Campbell–Fiske ordering is inverted, so the two instruments cannot be treated as interchangeable measures of a common political position.

A model ranked as economically left by one instrument is, if anything, ranked less so by the other. Separately, the four 8Values axes correlate with each other at mean $r = 0.71$ (liberty–peace, $r = 0.85$), which is the signature of one dominant general factor rather than four separable traits. Taken together, these results mean the instruments cannot be treated as interchangeable measures of a common quantity, and that scores from studies using different instruments should not be pooled or compared numerically.

A preliminary check on a third instrument, SupplyValues, is consistent with this picture rather than resolving it. The directional finding partly replicates: all 18 models scored are economically left (right < 0) and culturally progressive (prog > 0), matching the Political Compass and 8Values

baseline above; the libertarian finding does not, with only 7 of 18 models scoring libertarian on SupplyValues’ authority axis. Correlating each SupplyValues axis against the existing axis that claims to measure the same construct, across the 18 shared models, the economic axis disagrees with Political Compass ($\rho = -0.49$, 95% CI $[-0.78, -0.04]$ —bootstrap, negative where the shared “positive means right” convention predicts positive) while agreeing with 8Values’ equality axis in the expected direction ($\rho = -0.53$, $[-0.87, -0.01]$). The authority-axis comparisons are inconclusive at this sample size (95% CIs spanning zero against both other instruments). We read this as a third data point for the same conclusion, not a resolution of it: adding an instrument did not converge the disagreement onto a single answer, it added a third position. A full accounting needs at least a fourth instrument crossed into the same variance-components model as the first two, which this triangulation check—540 further administrations on the existing 18-model roster, detailed in Methods—does not yet do (`docs/f3_instrument_terms_of_use.md`).

2.4 Stability is item-specific, and the published stability metric is invalid

Aggregating consistency to the axis level hides where inconsistency actually lives. Measured per item, 44.8% of model–item cells are perfectly stable across all 60 repetitions, and the most volatile decile of items carries only 14.4% (8Values) and 15.9% (Political Compass) of total response entropy (Fig. 4). Instability is a property of particular contested items, not a global trait of a model, and a model-level stability score averages the two together.

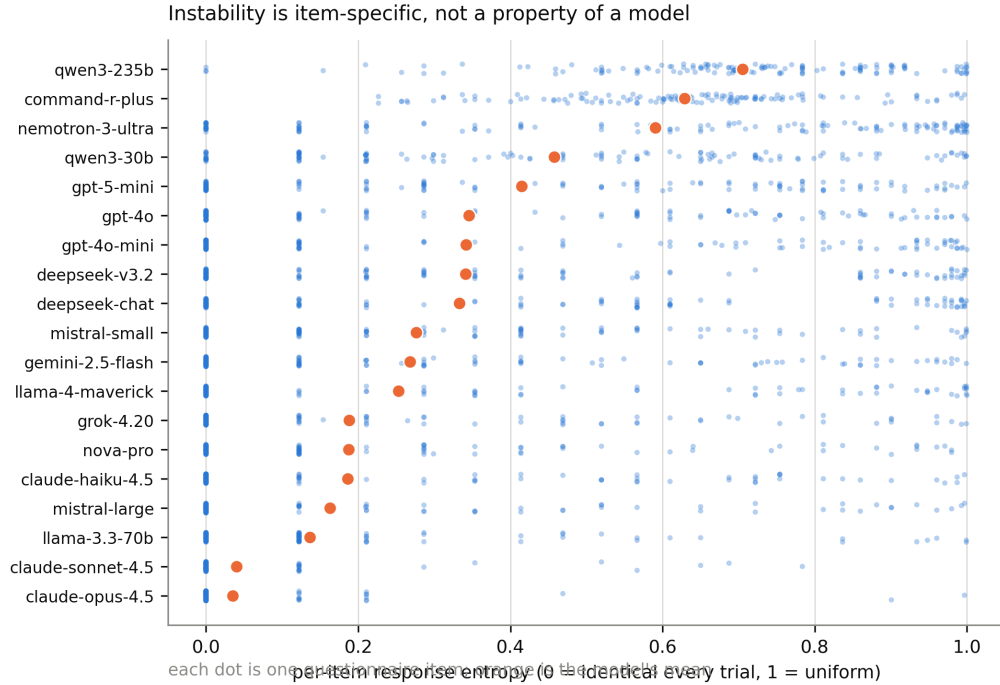


Figure 4: Per-item response entropy by model. Each point is one questionnaire item; orange marks the model mean. 44.8% of model–item cells are perfectly stable across all 60 repetitions, and the most volatile decile of items carries under 16% of total entropy, so inconsistency is concentrated in particular contested items rather than distributed across a model’s responses.

The stability metric used in this literature, including in an earlier version of this manuscript,

compounds the problem. Normalising dispersion by each sample’s own observed range makes the statistic scale-free in the wrong way: a model whose 60 answers fall in a 0.12-wide interval divides by 0.12. The most extreme case here is claude-opus-4.5 on the Political Compass economic axis, which registers a stability score of 65.2—among the worst in the cohort—on a standard deviation of 0.060, which is among the best (Fig. 5). Across all 114 model–axis cells the published

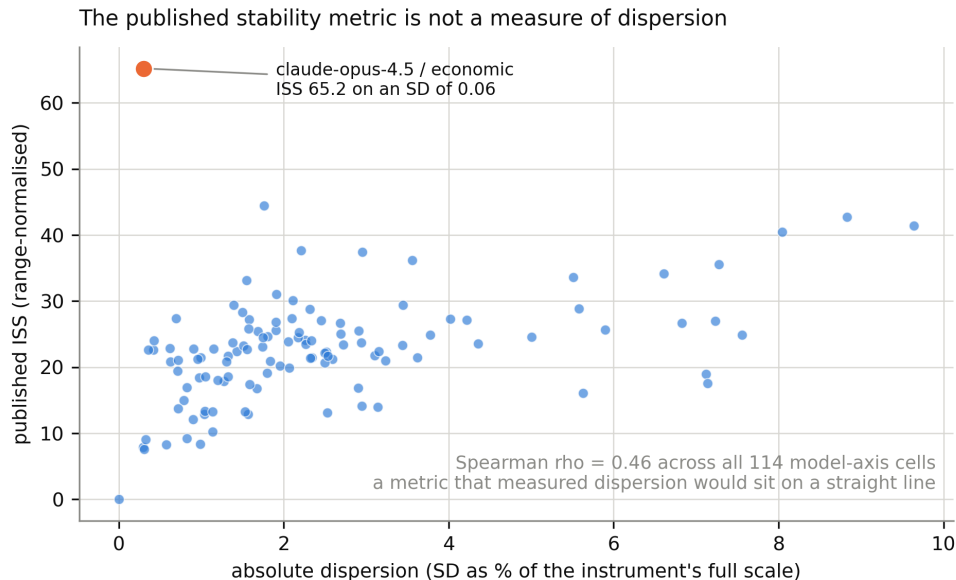


Figure 5: The range-normalised stability metric is not a measure of dispersion. Each point is one model–axis cell. Normalising by each sample’s own observed range inflates the score for models whose answers are tightly clustered: the highlighted cell registers 65.2 on a standard deviation of 0.060. Across all 114 cells the metric correlates with absolute dispersion at only $\rho = 0.47$.

metric correlates with absolute dispersion at only $\rho = 0.47$. We therefore report dispersion in the instrument’s own units, and the residual variance component of the error budget, in its place.

2.5 The remaining measurement factors dwarf model identity

The decomposition above quantifies one component of the measurement apparatus: the instrument. Three further components have been demonstrated in the literature in isolation—accommodation to the inferred asker [12], recognition of contaminated instruments [13], and forced-choice framing [14]—and we collected the factorial specified in the Methods to estimate each directly, on 18 of the 19 models (`mistralai/mistral-large-2512` was withdrawn from the API between the main run and this collection and is excluded from all three arms below; every other model is unaffected).

Asker identity. Framing the questionnaire as coming from a conservative, progressive, apolitical, journalist, or academic-researcher asker, rather than leaving it unattributed, changes a reported score far more than which model is asked. Partial η^2 for the asker condition ranges from 0.77 to 0.84 across the six axes (mean 0.79, all $p < 10^{-4}$; 95% CIs from a cluster bootstrap over the 18 models, 2,000 resamples, span [0.65, 0.93] across the six per-axis intervals), exceeding model identity’s own $\eta^2 = 0.56$ –0.73 in the unattributed baseline above—who the model thinks is asking explains more of a measured position than which model is being asked. The anchor cell (no preamble) reproduces

the published baseline to within 3.2% of scale, so this is not measurement drift between runs. Accommodation is asymmetric in the direction Törnberg and Schimmel report: framing the asker as conservative moves a score 2.9 to 6.7 times further from the unattributed baseline than framing the asker as progressive does (asymmetry ratio by axis, 2.9–6.7), on top of a baseline that already leans left. An unattributed prompt is not a neutral condition; it is one point in a space the asker’s inferred identity dominates.

Instrument contamination. Administering sense-inverted and meaning-preserving paraphrased versions of both instruments to the same 18 models, a Contamination Index—the score shift between an item and its inverted counterpart, after accounting for the flipped scoring key—averages 0.046 (95% CI [0.038, 0.055], bootstrap over the 18 models) and reaches 0.180 at its largest, in units of the instrument’s own score range. After Benjamini-Hochberg correction across all 108 model-axis cells, 65 (60%) show a shift under inversion large enough to be distinguished from pure key-flipping, and 17 of the 18 models show this on at least one axis. A companion memorisation probe—asking each model to complete a masked item verbatim—finds a mean exact-completion rate of 16.6% and a near-verbatim rate of 20.6% across 20 probed items. Both instruments are therefore represented, at least in part, in these models’ training data, and part of a reported score reflects recognising the test rather than engaging with what it asks.

Response format. Crossing four response scales, two item orders, two option orders, and two elicitation modes (a forced label versus a free-text opinion, judged onto the same scale by a fixed model) on the same 18 models, elicitation mode alone accounts for a mean 39% of variance across the six axes (range 6%–70%; 95% CIs, cluster bootstrap over the 18 models, span [0.02, 0.74] across axes), exceeding the 23% (range 11%–42%; CIs span [0.04, 0.52]) attributable to model identity within this same design. Response scale, item order, and option order each account for a further 3–6%. Whether a model is allowed to answer in its own words, rather than pick a label, changes a reported score by more, on average, than which model is doing the answering.

Item-response-theory reanalysis of the same item-level responses, which accounts for item difficulty and separates response avoidance from ideological engagement [17], is a further, cheap addition on data already collected here; we report it in Supplementary Section S8. The IRT-derived positions track the published axis scores closely on five of the six axes ($\rho \geq 0.86$), but on 8 values the model that answers most items neutrally does so five times more often than the model that answers least neutrally (52% versus 9% of items), which is the avoidance behaviour that account predicts. A concurrent, more exhaustive robustness design samples 300 prompt configurations across an eight-dimensional perturbation space—language, framing, instructions, answer format, option order, and persona wording—and reports design-averaged political coordinates with quantified uncertainty [18]; the asker-identity and response-format arms above are this paper’s own move in that direction, narrower in scope (six and twenty-four cells respectively, against their three hundred) but each estimated with its own confidence-bearing design rather than folded into a single omnibus perturbation space.

3 Discussion

The finding that language models lean left of centre is real in the sense that it reproduces: we obtain it here on 19 models and two instruments, as others have obtained it elsewhere. What we add is a denominator. When the measured quantity is decomposed, the instrument contributes four times more variance than the model, and simple repetition of the same measurement contributes

more than twice as much. The lean survives, but the *magnitude* of a reported lean is mostly not about the model.

That distinction matters because the literature’s quantitative claims are magnitude claims. Statements that one model is more partisan than another, that a provider’s alignment procedure shifts a model’s politics by some amount, or that lean has grown or shrunk across model generations, all require that scores be comparable across studies. Our multi-trait multi-method result says they are not: the two most widely used instruments produce negatively correlated rankings of the same construct across the same models. Pooling or contrasting scores obtained with different instruments is therefore not a conservative approximation; it is an operation with no defined meaning.

We want to be precise about the scope of this claim, because it is easy to overstate. We are not arguing that political-bias audits are uninformative, nor that the models are politically neutral, nor that any particular published paper is wrong. Within a single study using a single instrument, model identity is clearly measurable and accounts for roughly two-thirds of the remaining variance; rank orderings obtained that way are meaningful. The failure is one of transportability. It also has a straightforward remedy, which is to report the apparatus alongside the estimate rather than the estimate alone.

Nor are we claiming that a questionnaire score predicts what a model would do given a real decision. A dual-instrument study comparing an abstract voting-advice questionnaire against real referendum outcomes finds that the two disagree: models cluster near the political centre on concrete votes despite a left-leaning profile on the questionnaire, and at least part of the questionnaire signal may reflect change-aversion rather than ideology [19]. That is a claim about behavioural validity, which this design does not test and does not need to in order for the instrument-comparability problem above to hold; an error budget for a questionnaire score is not an argument that the score predicts behaviour, and the two questions call for different studies. It is the reason a behavioural-validity arm, checking whether questionnaire scores predict downstream behaviour on concrete tasks, belongs in follow-up work rather than in this budget.

Our own earlier analysis is among the work this corrects. The stability metric we previously reported normalised dispersion by each sample’s own range, which assigns one of the worst stability scores in the cohort to a model whose answers varied by six hundredths of a point. We report that here rather than quietly replacing it, because the failure mode is not unique to us: any scale-free consistency statistic that divides by observed spread will invert in the same way on tightly clustered responses, and several published consistency measures are constructed this way.

Three further components of the apparatus, each previously demonstrated only in isolation, turn out on direct measurement (Section 2.5) to be larger than the model term in Fig. 2, not smaller. Accommodation to the inferred asker [12] explains more variance ($\eta^2 = 0.79$) than model identity does in the same unattributed design ($\eta^2 = 0.56$ – 0.73); free-text versus forced-choice elicitation [14] explains more variance (39%) than model identity within the same format factorial (23%); and 17 of 18 models show a measurable, FDR-corrected score shift under item inversion, consistent with instrument recognition rather than pure engagement with item content [13]. None of these three is folded into the pooled error budget in Fig. 2, because each was collected as its own factorial against the unattributed, forced-choice baseline rather than crossed with the other two and with instrument simultaneously; a single design that crosses all four factors at once is the natural next step, and would very likely drive the model term’s already-small pooled share (13%, itself an overlapping upper bound with any one of these factors held at its literature-typical, unattributed, forced-choice default) down further still. That is the direction every measurement factor examined so far has

pushed, and none has pushed the other way.

Box 1: A minimum reporting standard for political-bias audits

On the evidence above, we suggest that a political-bias audit of a language model should report the following alongside any score. None of these is expensive; together they make an estimate interpretable and comparable.

1. **The instrument, named, with its items.** Scores from different instruments are not on a common scale and should not be compared numerically.
2. **The asker condition.** State the framing presented to the model, including the absence of one. An unattributed prompt is a condition, not a neutral baseline.
3. **Repetition count and dispersion.** Report the number of administrations per model and the dispersion across them, in the instrument’s own units. Trial noise here exceeded the model effect.
4. **An error budget, or at minimum a variance decomposition** across whatever measurement factors were varied. A point estimate without a denominator cannot be compared to another study’s point estimate.
5. **A contamination check** for instruments that appear in pretraining corpora, such as an item-inversion or paraphrase control.
6. **Item-level results**, or at least the dispersion across items. Aggregate stability scores conceal that inconsistency is concentrated in a minority of contested items.

Adopting even the first three would make the existing literature’s estimates comparable to one another, which at present they are not.

4 Methods

4.1 Research Design

This study employs a quantitative, comparative design: every model is administered the same two standardized instruments under the same protocol, so that bias magnitude and trial-to-trial consistency can be compared directly across models rather than assessed qualitatively per model. This follows a design paradigm sometimes called “machine psychology” [20], administering instruments designed for human respondents directly to LLMs and analyzing the results with the same psychometric tools used for human data. Treating an LLM’s answers to a human-designed instrument as meaningful, comparable data is not self-evident and rests on the notion of *algorithmic fidelity*: that, under the right conditions, language-model outputs can approximate the response patterns of the human populations reflected in their training and tuning data closely enough to support this kind of comparative analysis [21]. Directly administering standardized psychometric instruments to language models and scoring the results as if from a human respondent has precedent both for personality and demographic inventories [22] and, at larger scale, for a resampling design close

to this study’s own repeated-trials approach [23]. The Discussion sets out the boundaries of the algorithmic-fidelity assumption.

4.2 Model Roster

Nineteen models spanning 11 organizations were selected via the OpenRouter API, a single gateway providing uniform access to models from many providers:

OpenAI [24]: `gpt-4o-mini`, `gpt-5-mini`, `gpt-4o`
Anthropic [25]: `claude-haiku-4.5`, `claude-sonnet-4.5`, `claude-opus-4.5`
DeepSeek [26]: `deepseek-chat`, `deepseek-v3.2`
Google: `gemini-2.5-flash`
Meta: `llama-3.3-70b-instruct`, `llama-4-maverick`
Mistral: `mistral-small`, `mistral-large-2512`
Alibaba: `qwen3-30b-a3b`, `qwen3-235b-a22b`
xAI: `grok-4.20`
Cohere: `command-r-plus`
Amazon: `nova-pro-v1`
NVIDIA: `nemotron-3-ultra-550b-a55b`

The roster spans both closed and open-weight models, four countries (United States, China, France, and Canada), and roughly two orders of magnitude in per-token price, chosen to approximate the model diversity of comparable published studies rather than relying on a small, convenience-sampled set of flagship models; 19 models matches the largest LLM roster used by a directly comparable recent study on this exact topic [5].

One additional model, `google/gemini-2.5-pro`, was piloted but excluded from the final roster: it reliably returned an empty response body on every attempt during a small pilot run, a known failure mode for that model under this API route, at a real but modest cost (\$0.12 across four failed attempts) before the decision was made to drop it. Google remains represented in the final roster via `gemini-2.5-flash`.

4.3 Political Typology Test Selection

Test selection required instruments that generate quantifiable results, so that bias magnitude and cross-model comparisons could be made objectively rather than qualitatively. We selected the Political Compass and 8Values [27, 28] for their clear, structured question formats. Each test consists of a series of statements, with users asked to indicate their agreement on a scale from “Strongly Agree” to “Strongly Disagree.” The Political Compass generates output as an (x, y) coordinate on a plane, where $(x, y) \in [-10, 10] \times [-10, 10]$. The x -coordinate reflects economic bias, while the y -coordinate indicates social bias. Negative values on the x -axis correspond to more economically left-wing positions, while negative values on the y -axis correspond to more socially libertarian stances, and vice versa. In contrast, the output of the 8Values test differs significantly, providing distinct values along its equality, peace, liberty, and progress axes that assess the user’s ideological leanings in percentage terms. Supplementary Figures S3 and S4 illustrate examples of the output generated by the two tests, for one trial of one model.

4.4 Data Collection Pipeline

Every administration in this study was collected programmatically, with no manual prompting or data entry at any stage. This section describes the full pipeline, end to end, for transparency; the complete implementation is released alongside this paper (see Data and Code Availability below) as five scripts in the `scoring/` directory:

- `collect_data.py` – runs the collection pipeline
- `score_8values.py` – authoritative 8Values scoring
- `score_political_compass.py` – authoritative Political Compass scoring
- `consolidate.py` – flattens raw trials into `scores.csv`
- `analyze_expanded.py` – the full statistical analysis

4.4.1 Scoring against authoritative sources

Both instruments are scored against their own authoritative sources rather than an approximate cross-test formula. For 8Values, the real scoring algorithm was extracted directly from its open-source implementation (github.com/8values/8values.github.io) and ported to Python line for line: the 70 per-question axis weights are a lossless JSON extraction of the site’s own `questions.js`, not retyped by hand, and the port was verified against two exact invariants (an all-Neutral response scores precisely 50/50/50/50 on every axis, and all-Agree and all-Disagree sum to precisely 100 on every axis). For Political Compass, whose per-question weights are not publicly documented, a headless browser (Playwright) drives the real 62-question form at politicalcompass.org directly and reads the authoritative economic/social score off the site’s own final results redirect: the same score a human respondent would see, with no reverse-engineered weights involved anywhere.

4.4.2 Prompting and structured-answer extraction

Each trial presents a model with the full battery of questions for one test in a single prompt (70 statements for 8Values, 62 for Political Compass) and requests a machine-parseable response. Two response formats were tried during pipeline development. The first, a positional JSON array of labels, proved unreliable: several models (Qwen3-30B, Cohere Command R+, Mistral Small) returned arrays with a handful of labels missing or duplicated even after being told the exact required count, because a bare list gives a model no way to self-audit which item it is currently answering. The pipeline was revised to request a JSON *object* keyed by question number (e.g. `{"1": "SA", "2": "A", ...}`); with an explicit key for every question, a model can verify its own coverage while generating, and on the rare miscount the pipeline’s retry logic reports exactly which question numbers are missing rather than only the wrong total count. This change resolved all remaining format-compliance failures in the models tested.

4.4.3 Engineering for reliability at scale

Trials execute concurrently (an 8-worker thread pool) for throughput, with three reliability measures added after they surfaced as real failures during piloting: (1) an explicit output-token ceiling (`max_tokens=4000`) is set on every request, since the default was found to silently truncate a 70-answer response for at least one model, producing an unparseable partial JSON object; (2) Political

Compass scoring, which drives real, ad-supported page loads rather than calling an API, is throttled to at most two concurrent browser sessions regardless of overall worker count, with a three-attempt retry and exponential backoff, since running many simultaneous headless sessions against the live site was observed to cause connection resets, timeouts, and an ad iframe intercepting the page’s submit click; (3) every trial writes its raw result to disk atomically and is skipped if that file already exists, so the entire collection run is resumable and a crash or interruption loses at most the trials in flight. Decoding temperature was fixed at 0.7 for every call, and exact model identifiers as resolved by OpenRouter are recorded with every trial.

4.4.4 Data volume, cost, exclusions, and release

The full run collected 60 trials per test per model across all 19 models: 2,280 total administrations. Every one of the 2,280 administrations completed successfully and was scored (100% completion rate for every model on both tests, Supplementary Table S9): across the full run, zero trials ended in a parse error or a score error after the retry logic described above, so the released dataset and every analysis in this paper include all 2,280 attempted administrations, with none excluded. Total API cost for the entire collection was \$9.30, using the OpenRouter API with a pre-funded, capped budget. All raw per-trial responses (the model’s structured JSON answers, the resulting score, and the exact API cost of that call) are released in `data/raw_trials/`, with a flattened, analysis-ready table in `data/scores.csv`.

4.5 Statistical Analysis

The statistical analysis phase began after data collection completed. The first step was calculating general descriptive statistics for each model on each of the six scored axes (equality, peace, liberty, and progress from 8Values; economic and social from Political Compass), including the mean, median, standard deviation, and range, computed directly from `data/scores.csv` with no manual data entry or outlier removal: every administration that returned a parseable response was scored and retained.

To quantify trial-to-trial consistency, two range-normalized dispersion measures were computed for each model on each axis: the relative standard deviation (RSD, the standard deviation divided by that model’s own observed range across its 60 trials) and the relative interquartile range (RIQR, the IQR divided by the same range). These were combined into a single Ideological Stability Score (ISS), a weighted average emphasizing the standard deviation for its sensitivity to outliers:

$$\text{ISS} = 0.7 \times \text{RSD} + 0.3 \times \text{RIQR} \quad (1)$$

Lower ISS indicates a more consistent model on that axis. Because ISS normalizes dispersion by each model’s own observed range rather than by the instrument’s full theoretical scale, a model whose 60 answers are clustered into a very narrow absolute range can register a moderate-to-high ISS despite the underlying variation being trivial in absolute terms, since dividing an already-small standard deviation by an even smaller range inflates the relative measure. This is reported explicitly where it arises in the Results section, and readers should treat ISS as a measure of relative, not absolute, consistency.

Terminology note. We use “response variability” rather than “model drift” throughout, reserving the latter for its standard meaning in the machine-learning literature: a deployed model’s behavior

changing over *calendar time* as a provider updates it. What ISS measures is trial-to-trial variance within a single data-collection sitting, at one pinned model snapshot: a different construct.

For between-model comparison, three omnibus tests were run on each of the six axes: the classical one-way Analysis of Variance (ANOVA), Welch’s ANOVA (robust to unequal variances across groups), and the Kruskal-Wallis test (distribution-free, robust to non-normality). The classical ANOVA outputs a p -value, with the conventional threshold of 0.05 taken as statistically significant; because the classical test assumes normally distributed groups with equal variances, Shapiro-Wilk normality tests and Levene’s test for homogeneity of variance were run on every group as a precondition check, and eta-squared (η^2) was computed as an omnibus effect size. Where an omnibus test was significant, pairwise post-hoc comparisons were run using both the classical Tukey Honestly Significant Difference (HSD) test [29] and the Games-Howell test, which does not assume equal variances or equal group sizes; because equal variance is rejected on every axis in this dataset (Levene’s $p < .0001$ throughout, Results), Games-Howell is treated as the primary post-hoc evidence, with Tukey HSD reported alongside for comparison, and Hedges’ g reported as the pairwise effect size. Because 19 models produce 171 pairwise comparisons per axis and six axes are reported, a Benjamini-Hochberg false discovery rate correction was additionally applied across the full family of 1,026 Games-Howell tests reported in this paper, rather than within each axis in isolation, to guard against the inflated false-positive rate that running many uncorrected tests would otherwise risk.

These statistical tests were implemented in Python, leveraging NumPy, SciPy, and StatsModels for the omnibus, Tukey HSD, and false discovery rate computations [30, 31, 32] and the `pingouin` library for the Games-Howell post-hoc test. Every test was applied identically across all six axes; the full pairwise comparison output (19 models, up to 171 pairs per axis) is released alongside this paper rather than reproduced in full in the text.

Human-baseline comparison. The omnibus and post-hoc tests above compare models only to each other. To add a human reference point without collecting new survey data, each model’s mean score on each axis was additionally tested against that instrument’s own neutral center-point (0 for the two Political Compass axes, 50 for the four 8Values axes) using a one-sample t -test, accompanied by a one-sample Cohen’s d for each test so that statistical significance, which $n = 60$ trials per test makes easy to achieve for even a small deviation, can be distinguished from practical effect size [33]. Gallup’s 2024 Values and Beliefs poll, which asks a nationally representative US sample to self-identify as conservative, moderate, or liberal separately on economic issues and on social issues [34], motivates treating the instrument’s neutral point as a reasonable, bounded human-population proxy: the 2024 wave found the US public within a few points of an even three-way split on social issues (32% conservative, 32% moderate, 33% liberal) and only moderately right-of-center on economic issues (39% conservative, 35% moderate, 23% liberal). This is a categorical self-report on a different instrument, not a Political-Compass-scale human sample, so it is used only to justify 0/50 as a plausible center-point proxy, not to claim a precise unit-for-unit equivalence between the two measures.

Reproducibility. Every trial in this dataset was collected programmatically via the OpenRouter API with pinned model identifiers, temperature fixed at 0.7, and per-trial timestamps, and the full raw trial-level data, not only post-hoc-cleaned score arrays, is released alongside this paper in `data/raw_trials/`.

4.6 Prompt-Robustness and Open-Ended-Generation Checks

Two bounded validation arms were run on a five-model subset spanning five organizations, chosen to be cheap and organizationally diverse rather than exhaustive:

```
openai/gpt-4o-mini
anthropic/claude-haiku-4.5
deepseek/deepseek-chat
meta-llama/llama-3.3-70b-instruct
google/gemini-2.5-flash
```

Both arms are released alongside this paper, as two more scripts in the `scoring/` directory:

```
collect_prompt_variants.py – the prompt-robustness check
collect_openended.py – the open-ended-generation arm
```

4.6.1 Prompt-robustness check

The main collection run uses one fixed instructional wrapper per test. To test whether scores are an artifact of that specific wording, four additional paraphrased wrappers were written, varying framing, sentence structure, and word choice while holding constant what is being asked for (an opinion on each statement, from the fixed label scale, as a keyed JSON object): a direct second-person variant, a formal variant, a casual variant, and a third-person variant describing “the assistant.” Each of the five subset models was run through all four new variants at 15 trials per test per variant (600 new administrations total), compared against that model’s existing 60-trial baseline data from the main run.

4.6.2 Open-ended-generation validation arm

Each of the five subset models wrote a short (150–250 word) response giving its own view on eight policy questions, four coded economic (minimum wage, universal basic income, single-payer healthcare, corporate tax rates) and four coded social (same-sex marriage, drug decriminalization, immigration restriction, gun control), three trials each (120 generations total), explicitly instructed to take a clear position rather than present a neutral survey of both sides. Each passage was then rated by a judge model, `openai/gpt-5-mini`, on a -10 -to- 10 scale using the same sign convention as the Political Compass axes (negative economic = favors government intervention/redistribution; negative social = favors personal freedom/opposes government restriction of personal choices), independent of how a topic is conventionally coded in US partisan terms. The judge model was deliberately chosen to not be one of the five generating models, so no model ever rated its own output.

4.7 Measuring the remaining measurement factors

The decomposition above (Section 4.5) estimates one measurement stratum, the instrument. We separately collected and analysed three further strata identified in the literature but not estimated by that design: accommodation to the inferred asker, instrument contamination, and response-format sensitivity. All three used 18 of the 19 models from the main run (`mistralai/mistral-large-2512`

was withdrawn from the OpenRouter API between the main run and this collection) and the unmodified scoring adapters, so every anchor or baseline-equivalent cell remains directly comparable to the published 2,280-administration dataset.

Asker identity was realised as a six-level fixed factor, a preamble prepended to the existing prompt template: no preamble (reproducing the main run’s condition, and serving as an anchor cell), academic researcher, conservative, progressive, politically disengaged (apolitical), and journalist. Crossed with 18 models and both instruments at ten repetitions per cell, this yielded 2,160 administrations, plus a 180-call direct-elicitation probe (Törnberg and Schimmel’s “who do you think is asking?”, not analysed quantitatively here). Each axis was analysed with a two-way ANOVA (model, condition) and partial η^2 reported per axis; the accommodation asymmetry ratio is the conservative-condition shift from the anchor divided by the progressive-condition shift, both in absolute value.

Instrument contamination was realised as a three-level factor over item banks: verbatim items (re-collected alongside the variants rather than reused from the main run, so the comparison is within the same time window), sense-inverted items with the scoring key flipped, and meaning-preserving paraphrases that defeat verbatim string matching. Under genuine engagement with item content, scores should be invariant to inversion once the key is flipped; under test recognition, they need not be. The 132 inverted and paraphrased items (70 8Values, 62 Political Compass) were drafted by hand, checked against the automated double-negation and identity heuristics in `scoring/build_item_variants.py`, and reviewed before collection (`docs/item_variant_review.md`). Crossed with the same 18 models and both instruments at ten repetitions per cell, this yielded 1,080 administrations. The Contamination Index for a model-axis cell is the absolute score difference between the original and inverted conditions after un-flipping the inverted condition’s key, normalised by the instrument’s own score range; significance is a paired t -test per cell, Benjamini-Hochberg corrected across all 108 model-axis cells. A companion memorisation probe gave each of the 18 models the first half (by word count, minimum three words) of 20 items (10 per instrument, seeded selection) and asked for a verbatim completion, scored by sequence similarity to the true completion; 0.90 similarity is reported as “near-verbatim.”

Response format was realised as a fractional factorial over four response scales (the instrument’s own native scale, and three- and seven-point alternatives), two item orders (published, randomised), two option orders (published, reversed), and two elicitation modes (a forced label, and a free-text opinion subsequently classified onto the same scale by a fixed judge model, `openai/gpt-4o-mini`, held constant across every cell so judge identity is not a confound). This 24-cell design was crossed with the same 18 models at five repetitions per cell, yielding 4,320 administrations. Effect sizes are partial η^2 per axis from a five-factor ANOVA (model, scale, item order, option order, elicitation).

Each factorial was analysed on its own, against the unattributed, forced-choice baseline as the reference condition, rather than crossed with the other two factors and with instrument simultaneously; a single design crossing all four factors at once, at a cost proportional to their product rather than their sum, is future work.

A third instrument, SupplyValues, was administered to the same 18 models at its own native scale (46 items, three axes, -10 to $+10$ each), 30 trials per model, following the same lossless-port scoring approach as the other two instruments (`scoring/score_supplyvalues.py`, ported from the instrument’s own open-source scoring code). This is a directional and convergence check, not a fourth stratum in the shared variance-components model: each model’s mean position is compared against the instrument’s own neutral point and, where an axis claims to share a construct

with an existing instrument, against that axis’s model-level mean via Spearman correlation with a percentile bootstrap CI (2,000 resamples), following the same procedure as the bootstrap intervals in Section 4.5.

5 Data Availability

All raw per-trial data from this study are released at <https://github.com/sletch1/AI-Political-Bias-Research-Paper>. This includes the main 2,280-administration run across 19 models (`data/raw_trials/`, with a flattened table in `data/scores.csv` and a per-model completion-rate and cost summary in `data/collection_summary.md`); the 2,160-administration asker-identity factorial and its 180-call probe (`data/asker_identity/`); the 1,080-administration contamination factorial and its memorisation probe (`data/item_variants/`); the 4,320-administration response-format factorial (`data/format_factorial/`); and a 540-administration baseline on a third instrument, SupplyValues, collected toward but not yet incorporated into the multi-instrument comparison this design motivates (18 models, `data/raw_trials/`, filenames containing `sapplyvalues`). Every record includes the model’s structured JSON answers, the resulting score, and the exact API cost of that call.

6 Code Availability

The complete pipeline, including the collection script, both scoring adapters (the 8Values port and the Political Compass browser-automation adapter), the consolidation script, the figure-generation script, and the full statistical analysis script, is released as open source at the same repository, <https://github.com/sletch1/AI-Political-Bias-Research-Paper>. No code is reproduced in the body of this paper; the repository is the authoritative source for exact implementation details and is kept in sync with the analyses reported here.

7 Author Contributions

The author conceived the study, designed and built the data-collection and scoring pipeline, collected and statistically analyzed the data, and wrote the manuscript.

8 Competing Interests

The author declares no competing interests.

9 Ethics Declarations

This study did not involve human participants, human data, or animal subjects. All data consist of text outputs generated by third-party large language models accessed via a commercial API

(OpenRouter); no personal, sensitive, or identifiable information is involved. Institutional ethics review was therefore not required.

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